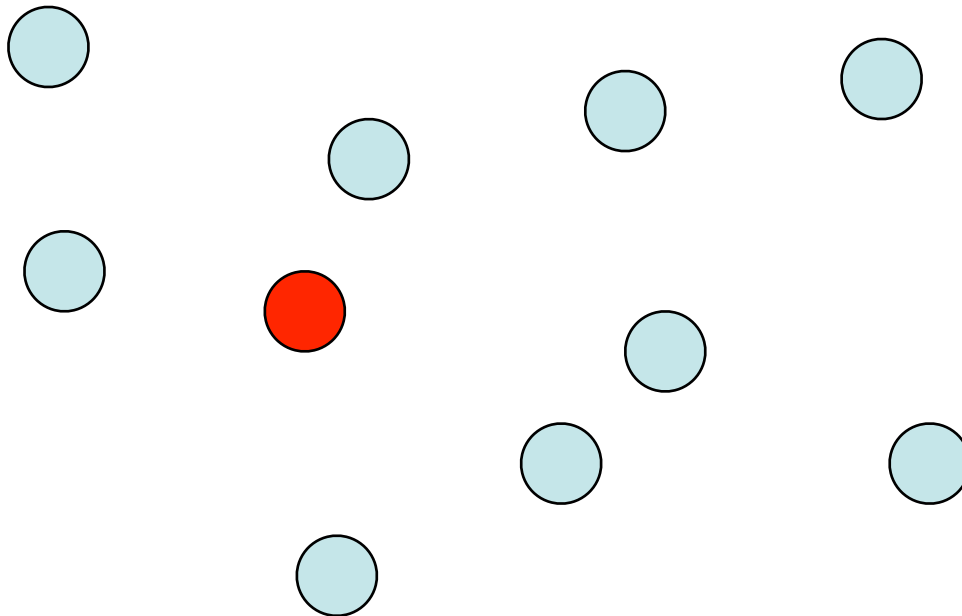


Epidemics, continued

Outline:

- (1) The SIR Model, revisited
- (2) Simulating the SIR model in MATLAB
- (3) Google Flu!
 - Reading Google flu data into MATLAB
 - Fitting Google Flu data with the SIR model
- (4) Google trends: trends as epidemics?

General Idea



3 populations: Susceptible
Infected
Recovered

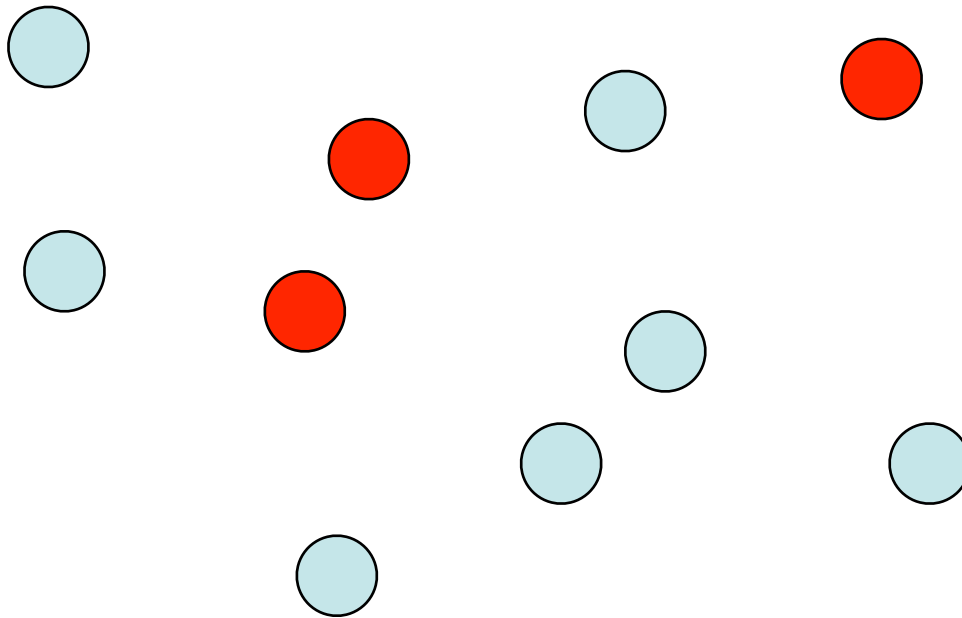
$S(t)$

$I(t)$

$R(t)$

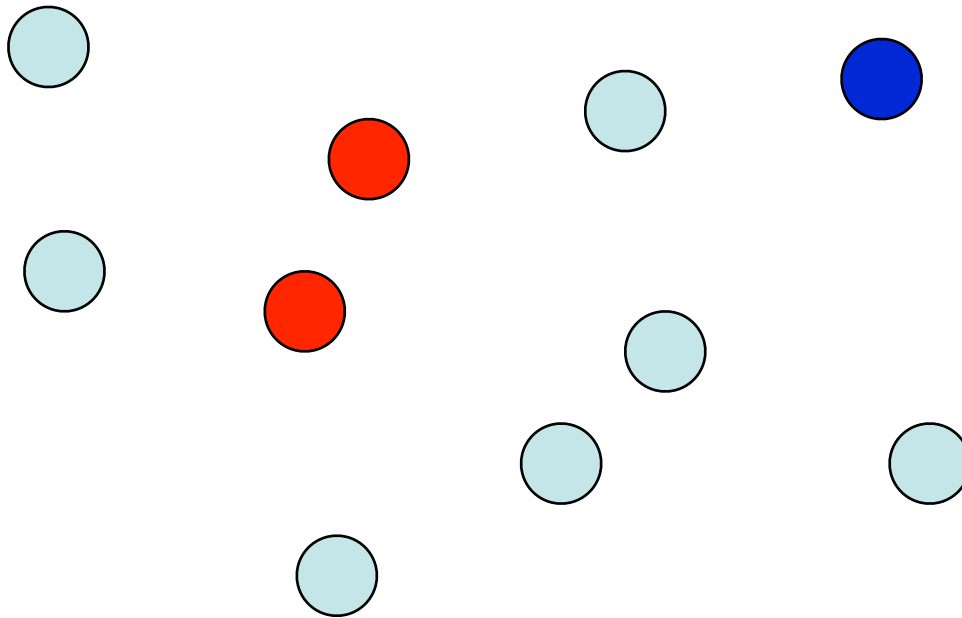


General Idea



3 populations: Susceptible $S(t)$ 
Infected $I(t)$ 
Recovered $R(t)$ 

General Idea



3 populations: Susceptible
Infected
Recovered

$S(t)$ 
 $I(t)$ 
 $R(t)$ 

$$\frac{dS}{dt} = -\beta IS$$



$$\frac{dI}{dt} = \beta IS - \gamma I$$



$$\frac{dR}{dt} = \gamma I$$

Two parameters

γ Rate of recovery/death of infected people

Units = 1/time

$$\frac{dI}{dt} = -\gamma I \quad \rightarrow \quad I(t) = I_0 e^{-\gamma t}$$

i.e. Suppose you recover in 1 week = 7 days

$$\gamma = \frac{1}{7} \quad \text{units of 1/days}$$

Two parameters

γ Rate of recovery/death of infected people

Units = 1/time

$$\frac{dI}{dt} = -\gamma I \quad \rightarrow \quad I(t) = I_0 e^{-\gamma t}$$

i.e. Suppose you recover in 1 week = 7 days

$$\gamma = \frac{1}{7} \quad \text{units of 1/days}$$

$$\gamma = \frac{1}{7} \times \left(\frac{1 \text{ day}}{24 \text{ hours}} \right) = \frac{1}{7 \times 24} \quad \text{units of 1/hours}$$

Two parameters

β

Rate of infected person infecting a susceptible person

Eg. in one hour $1/N$ will get sick if infected person is in the room

$$\beta = \frac{1}{N}$$

units of 1/hours

An amazing Conclusion

$$\frac{dI}{dt} = \beta IS - \gamma I = I(\beta S - \gamma)$$

$$S > \frac{\gamma}{\beta}$$

Epidemic spreads!

$$= \frac{N}{168}$$

How can you prevent flu epidemic?

$$S > \frac{\gamma}{\beta}$$

Epidemic spreads!

(1) Decrease beta

(2) Treat more effectively (or quarantine people who are sick!) -- increases gamma!

(3) Change population size of Susceptibles!
Vaccination!

Public health strategies are completely connected to this mathematical model.

Qualitative properties

$$\frac{dS}{dt} = -\beta IS$$

$$\frac{dI}{dt} = I \left(\beta S - \gamma \right)$$

$$\frac{dR}{dt} = \gamma I$$

Solve with MATLAB

To solve ordinary differential equations, need to specify:

- (1) The *initial conditions* $S(t=0)$, $I(t=0)$, $R(t=0)$
- (2) How long you want to integrate for , i.e. $0 < t < 100$ (hours!)

From the Lab

Script

```
tspan = [0 2];      % specifies the time span and initial condition
x0 = 0;
[t, x] = ode23('simpleode', tspan, x0);
plot(t,x)           % plot x(t) vs t
xlabel('t');         % label axes
ylabel('x');
```

Function

```
%This function computes  $\dot{x} = x + t$ 
function xdot = simpleode(t,x)
xdot = x + t;
```

Discuss Solving SIR Model in MATLAB

Show the different behaviours.



Detecting influenza epidemics using search engine query data

Jeremy Ginsberg¹, Matthew H. Mohebbi¹, Rajan S. Patel¹, Lynnette Brammer², Mark S. Smolinski¹ & Larry Brilliant¹

¹Google Inc. ²Centers for Disease Control and Prevention

We sought to develop a simple model which estimates the probability that a random physician visit in a particular region is related to an influenza-like illness (ILI); this is equivalent to the percentage of ILI-related physician visits. A single explanatory variable was used: the probability that a random search query submitted from the same region is ILI-related, as determined by an automated method described below. We fit a linear model using the log-odds of an ILI physician visit and the log-odds of an ILI-related search query:

$$\text{logit}(P) = \beta_0 + \beta_1 \times \text{logit}(Q) + \epsilon$$

where P is the percentage of ILI physician visits, Q is the ILI-related query fraction, β_0 is the intercept,

Publicly available historical data from the CDC's U.S. Influenza Sentinel Provider Surveillance Network¹² was used to help build our models. For each of the nine surveillance regions of the United States, CDC reported the average percentage of all outpatient visits to sentinel providers that were ILI-related on a weekly basis. No data were provided for weeks outside of the annual influenza season, and we excluded such dates from model fitting, though our model was used to generate unvalidated ILI estimates for these weeks.

Search Query Topic	Top 45 Queries		Next 55 Queries	
	N	Weighted	N	Weighted
Influenza Complication	11	18.15	5	3.40
Cold/Flu Remedy	8	5.05	6	5.03
General Influenza Symptoms	5	2.60	1	0.07
Term for Influenza	4	3.74	6	0.30
Specific Influenza Symptom	4	2.54	6	3.74
Symptoms of an Influenza Complication	4	2.21	2	0.92
Antibiotic Medication	3	6.23	3	3.17
General Influenza Remedies	2	0.18	1	0.32
Symptoms of a Related Disease	2	1.66	2	0.77
Antiviral Medication	1	0.39	1	0.74
Related Disease	1	6.66	3	3.77
Unrelated to Influenza	0	0.00	19	28.37
	45	49.40	55	50.60

Table 1: Topics found in search queries which were found to be most correlated with CDC ILI data. The top 45 queries were used in our final model; the next 55 queries are presented for comparison purposes. The number of queries in each topic is indicated, as well as query volume-weighted counts, reflecting the relative frequency of queries in each topic.

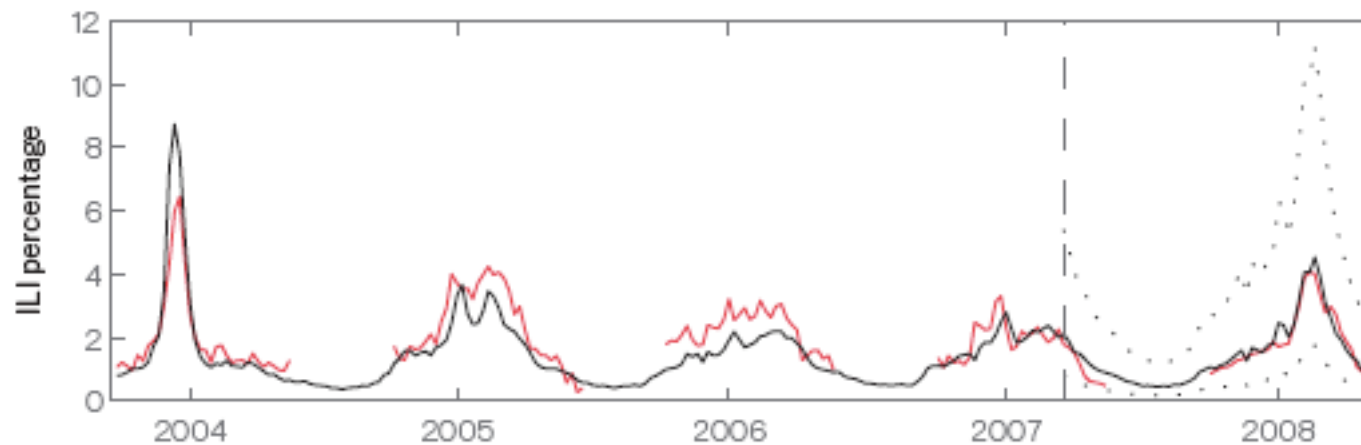


Figure 2: A comparison of model estimates for the Mid-Atlantic Region (black) against CDC-reported ILI percentages (red), including points over which the model was fit and validated. A correlation of 0.85 was obtained over 128 points from this region to which the model was fit, while a correlation of 0.96 was obtained over 42 validation points. 95% prediction intervals are indicated.

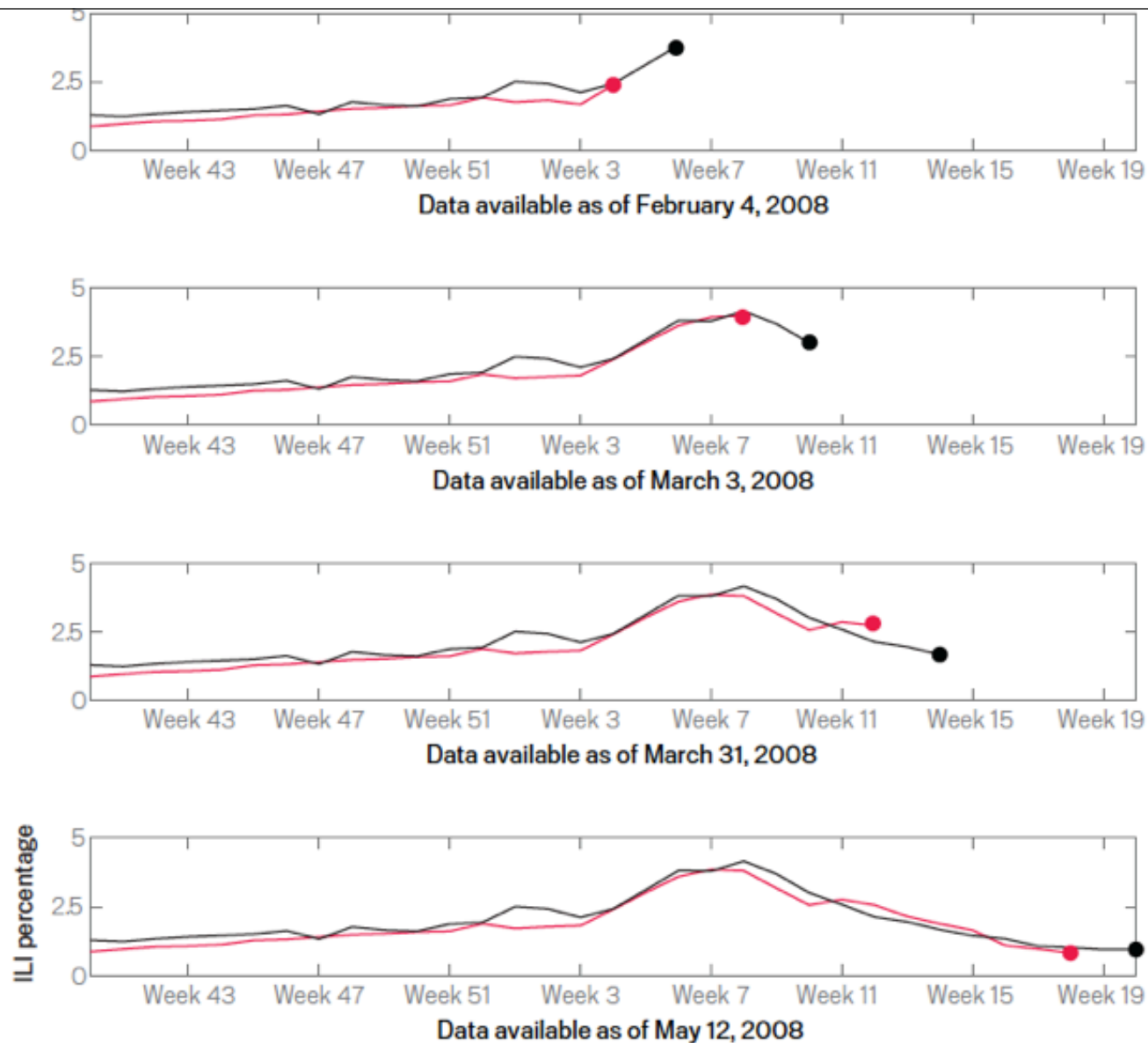


Figure 3: ILI percentages estimated by our model (black) and provided by CDC (red) in the Mid-Atlantic region, showing data available at four points in the 2007-2008 influenza season. During week 5, we detected a sharply increasing ILI percentage in the Mid-Atlantic region; similarly, on March 3, our model indicated that the peak ILI percentage had been reached during week

[Google.org home](#)

Flu Trends

Select country/region

Home

[How does this work?](#)

[FAQ](#)

Flu activity

Intense

High

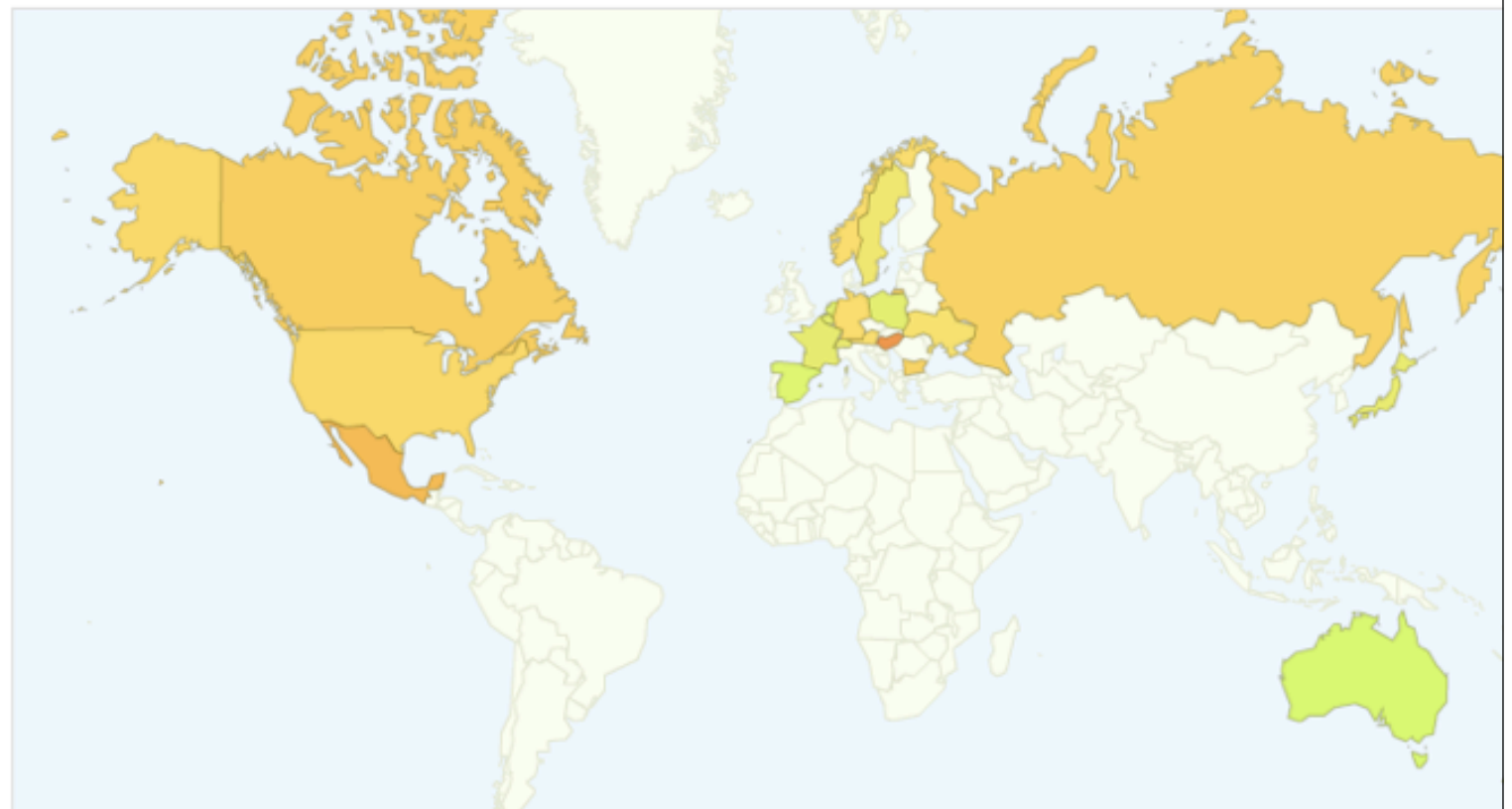
Moderate

Low

Minimal

Explore flu trends around the world

We've found that certain search terms are good indicators of flu activity. Google Flu Trends uses aggregated Google search data to estimate flu activity. [Learn more »](#)



[Google.org home](#)

Flu Trends

United States
National

[Download data](#)

[Home](#)

[How does this work?](#)

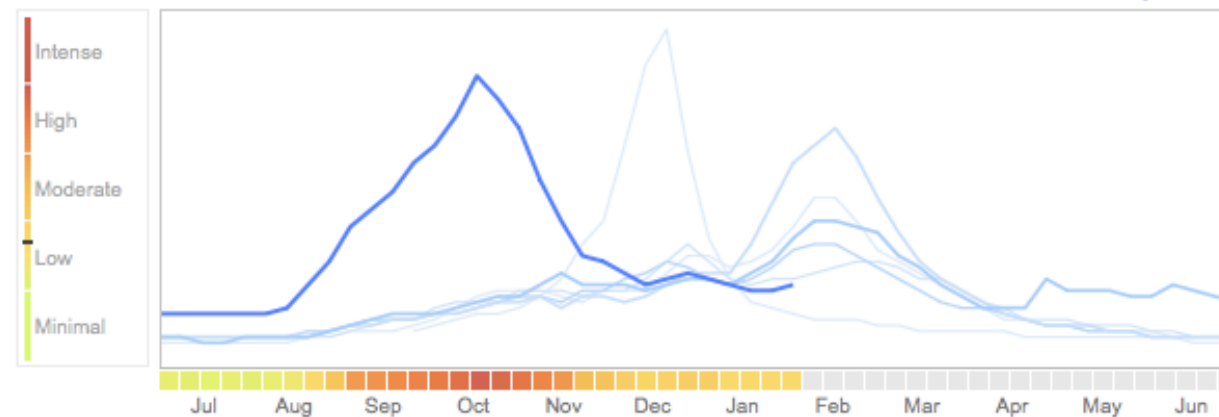
[FAQ](#)

Explore flu trends - United States

We've found that certain search terms are good indicators of flu activity. Google Flu Trends uses aggregated Google search data to estimate flu activity. [Learn more »](#)

National

● 2009-2010 ● Past years ▼



Fight influenza

CDC urges you to protect yourself and others from the flu:

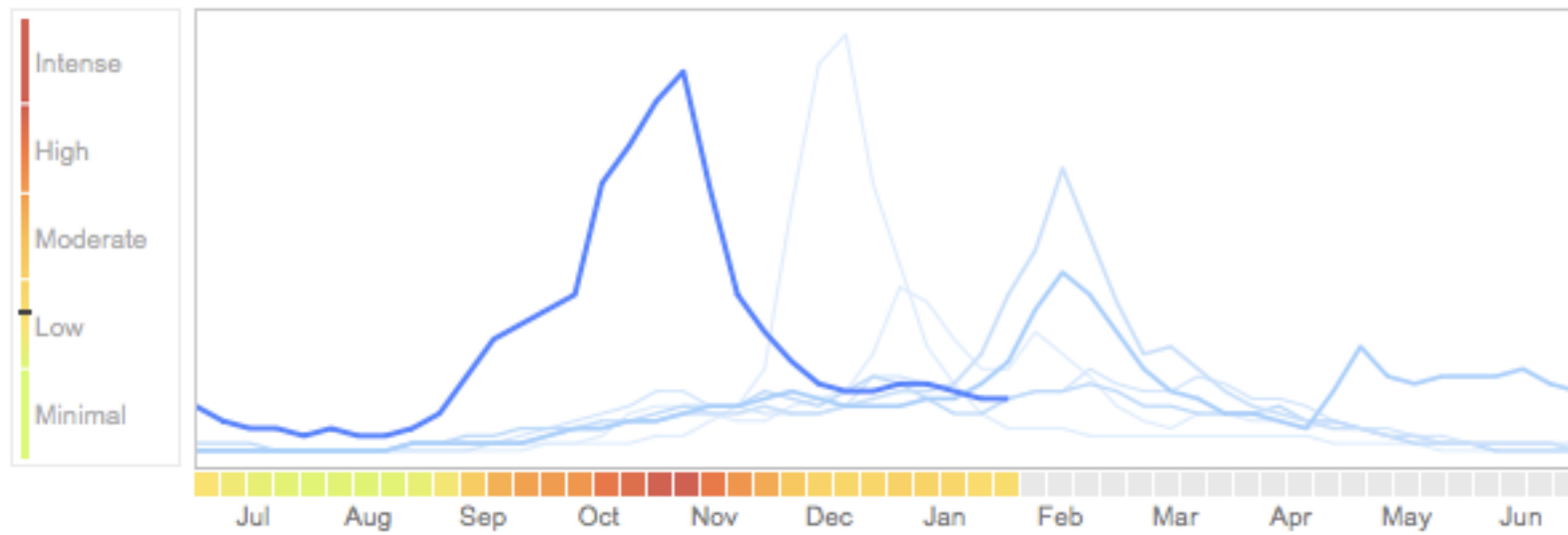
1. Get vaccinated
2. Wash hands often and avoid coughing or sneezing into your hands.
3. Take antiviral medications if recommended by your doctor.

This flu season could be different from prior years. A new H1N1 is spreading. Follow these 3 steps to protect yourself from the new flu, too.



United States > Boston, MA

● 2009-2010 ● Past years ▼



Questions for this week's lab

(1) Solve SIR model numerically

experiment with the different parameters and explore how large the epidemic is.

(2) Think about the role of vaccination.

What fraction of a population needs to be vaccinated to have it make a difference.

(3) Download data from Google flu.

Play around with SIR model parameters to try to fit the data.

(**) Questions to think about

Geographical spread of disease? Do the curves make sense in light of the relative parameters in the different states, countries, etc?

Google Trends...

Does the SIR model describe search query data in other cases?

Susceptible=population that might become interested in a topic

Infected=You are interested! You tell others about it and they get interested

Recovered=you are over it!